

ObserverBench: Testing Mechanistic Estimates for Intervention and Control

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Abstract

Mechanistic interpretability is increasingly used to guide interventions, such as steering activations, removing circuits, or enforcing safety boundaries. But an internal estimate that is accurate on average can still lead to a poor decision when it is used to choose a specific action.

We present **ObserverBench**, a workbench for testing whether an internal estimator—an observer—is adequate for the intervention, control, or safety task it directs. Each task fixes the model, information available to the observer, allowed actions, decision rule, unseen test cases, and loss. The task experiments report both estimation accuracy and downstream action performance, with reusable runners and data-only submission interfaces. Rankings are task-specific: observers are compared only under the same information and action constraints.

Our theory and experiments show three ways these criteria can differ:

- **Task-relevant accuracy.** In a control loop, observer errors matter at the starting point and along directions the allowed intervention can reach.
- **Prediction versus decision.** On circuit-intervention tasks in GPT-2-small and Qwen2.5-7B, pairwise observers predict unseen intervention effects more accurately without always choosing better actions. Observers trained to predict action loss directly choose lower-loss actions. On one Qwen APPS panel, a 127-coefficient SAE readout has higher mean audit loss than its layer-matched 3,584-coordinate dense control, under a disclosed activation-density mismatch (observed $L_0 = 115$ versus reported 240): compression without a better decision on that panel. Here compression means fewer readout coefficients, not lower total compute; this is separate from the prediction–action ranking reversals.
- **Consequence-aware safety.** Under fixed intervention budgets, a score that perfectly separates policy violations can allocate actions poorly when violations have different costs. On matched Qwen2.5-7B, Gemma-2-9B-it, and prospectively frozen Qwen3.5-9B APPS tasks, residual access leads on average, but the lowest-loss context changes—even between the two Qwen checkpoints—and AUROC can rank monitors differently from deployment loss. Gemma supplies the primary checkpoint-matched official-SAE comparison. The Qwen2.5 sparse probe and Qwen3.5 official SAE provide supporting checks under disclosed activation-density and checkpoint-transfer limits.

ObserverBench provides a common way to evaluate interpretability methods through the actions they enable.

1 Introduction

Interpretability methods are increasingly used to steer and govern model behavior rather than merely explain it. Internal estimates now guide concrete decisions: which activation-steering direction to apply, which attention heads to remove, whether to permit a risky tool call, or which requests receive a limited human-review budget. Activation engineering and feedback-control methods make these estimates part of an active control loop [3, 4, 5, 6]. Once an internal estimate is used to choose an action, statistical accuracy alone is not enough [29, 30, 15]: an estimate can perform well on average and still cause costly errors on individual prompts.

To analyze these systems, we separate the decision loop into three parts. The **observer** reads measurements and estimates an internal state or intervention effect. The **controller** uses that estimate to choose an action.

The **intervention** is the edit, ablation, or policy action it is allowed to take. ObserverBench asks whether the observer is adequate for that specific decision.

Three failure modes illustrate why prediction accuracy and decision quality must be reported separately. In safety triage, an observer can classify policy violations perfectly yet spend a limited review budget on low-stakes cases while allowing higher-impact harms through. This is *consequence disregard* (Sec. 5). In model editing, an observer can predict a circuit ablation’s mean effect accurately yet choose an edit whose effect varies widely across prompts. This is *dispersion blindness* (Sec. 4.1). An internal activation probe can also repeat information already visible in the prompt or output logits. Such a *redundant signal* adds model access without improving the downstream decision (Sec. 5).

We introduce **ObserverBench**¹, a set of fixed, reusable tasks that makes these operational distinctions measurable. Each task pins the model, information available to the observer, allowed interventions, controller, measurement budget, held-out test cases, and loss. Every observer faces the same controller and evaluation harness within a task. The task experiments report two outcomes separately: how accurately the observer predicts its target and how much loss its chosen action incurs on held-out cases.

ObserverBench focuses on the operational stage of interpretability. Mechanistic tomography addresses the measurement problem: how to design probes and interventions that recover hidden internal mechanisms and effect maps. ObserverBench addresses the decision problem: whether a proposed estimate is adequate to guide a particular intervention, control loop, or safety policy [1]. For safety tasks, ObserverBench freezes the audit policy and compares the monitor inside it. AI-control evaluations ask the complementary, harder question of whether the complete protocol remains safe when an adaptive model tries to subvert it [20]. A monitor that performs well here must still pass that end-to-end test.

The general distinction between prediction and decision quality is established in decision-focused learning and in traffic engineering [29, 30, 22]. Our contribution has three parts: a contract that freezes the decision problem and compares interchangeable observers within it; control conditions that identify where an observer must be correct; and empirical findings about intervention and safety decisions. These findings remain useful independently of the software. The workbench lets others test whether they persist with different observers.

An ObserverBench evaluation makes five choices explicit. It states the *target*: an internal state, an intervention’s average effect, or perhaps the loss of one action on one prompt. It states the *information and integrity boundary*: what the observer may read and who can alter, omit, delay, or spoof those measurements. It fixes the *action space* and the *decision rule and loss*, which determine how estimates become actions and which errors matter. Finally, it declares the *operating regime*, including prompts, intervention budgets, attack frequencies, episode horizons, and attacker assumptions.

These choices make observer adequacy task-relative: permissions, tool interfaces, budgets, attack rates, side channels, and evaluator-held evidence are part of the evaluation, not background details. In the 2026 OpenAI–Hugging Face incident, agents used an unsanctioned channel and spoofed some tool calls, motivating the integrity boundary [31, 32, 33]. ObserverBench instantiates these choices in three standardized operational modes—closed-loop control, effect prediction, and safety triage—summarized in Sec. 2.

Contributions.

- **An extensible evaluation workbench.** Versioned task and observer cards support local implementations and table-based submissions under declared task contracts. Control and safety interfaces score actions; the generic effect-table interface scores prediction error, while separate experiment runners evaluate the chosen intervention (Sec. 2).
- **A formal adequacy test for control.** For linear readouts with an offset, an observer needs to match the target only at the starting point and along directions the allowed interventions can reach. An exact finite-horizon identity then separates residual reported error, starting-state mismatch, and target–observer response mismatch along the edit direction (Sec. 3, Eq. 1).
- **Prediction and compression are not decisions.** Pairwise models improve held-out effect prediction on GPT-2-small and Qwen2.5-7B, but better mean prediction does not reliably improve the chosen GPT-2 action. Relative to the same-feature mean observer, direct-loss training reduces action loss by 18.5% in the clean GPT-2 confirmation and by 8.9% in the Qwen aggregate [16, 23, 24]. In the Qwen2.5 safety task,

¹Public workbench and submission page: <https://kwisatzh.github.io/observerbench/>.

a same-checkpoint sparse SAE readout uses far fewer coefficients yet incurs more escaped attacks under a disclosed activation-density mismatch: *compression without a better decision*. This is a readout-sparsity result, not a runtime saving or another reversal between prediction and action rankings (Secs. 4.1.5, 4.2, and 5) [28].

- **Classification is not enough for safety.** In the authorization task, a label observer with AUROC 1.000 allocates a fixed intervention budget poorly when it ignores consequence. On the external Gemma APPS task, AUROC and realized violations reverse the order of two monitors. Across matched Qwen2.5-7B, Gemma-2-9B-it, and prospectively frozen Qwen3.5-9B tasks, residual observers lead on average, while the lowest-loss context changes—even between the two Qwen checkpoints. Gemma supplies the primary checkpoint-matched official-SAE comparison; the Qwen SAE results are supporting tests with stated density or checkpoint-transfer limits (Sec. 5) [20, 21, 25, 26].

The next question is operational: what exactly does ObserverBench fix, what may a researcher submit, and what does the evaluator return?

2 How ObserverBench evaluates an observer

A fair comparison keeps the decision problem fixed. Otherwise, lower loss could reflect a better observer, a more capable controller, easier actions, or a different test distribution. ObserverBench therefore treats these choices as part of the task.

Each evaluation uses a **TaskCard**, an **ObserverCard**, and a **result scorecard** in four steps:

1. **Choose a task.** The TaskCard defines the decision problem using the five choices above and fixes the training measurements, held-out cases, budgets, and clean-performance checks. These do not change when observers are compared.
2. **Submit an observer.** Any method may be used if it respects the task’s information and integrity boundary. The ObserverCard records what the observer reads, where those measurements come from, how it was fit or calibrated, what model access it requires, and any known failure conditions. For a control task, it also declares whether the submission replaces the state estimate, the edit direction, or both. If both change, the score applies to the pair.
3. **Run the common evaluator.** Every observer receives the same held-out cases and faces the same controller, actions, budgets, and loss.
4. **Compare outcomes and cost.** The result scorecard reports prediction quality, downstream action loss, and the access or measurement cost of obtaining the estimate. The task determines which of these fields its submission interface currently returns.

There are two ways to cross the observer–evaluator boundary. A local implementation can fit and run a method that needs model inference or internal activations. A public submission can instead provide a table of predictions or scores for the task’s released cases. The second route lets a researcher test an observer without installing the original model or experiment stack. In both routes, the task—not the observer—holds the evaluation targets. Control and safety interfaces apply the frozen controller. The generic effect-table interface currently returns prediction metrics only; the intervention studies use separate task runners to select actions and measure their loss. Connecting that final step to arbitrary submitted tables is the next implementation task (Appendix A). Access declarations and readout sizes are reported where available; they are not end-to-end runtime measurements.

The three operational modes share this contract but ask different questions (Table 1).

Table 1: ObserverBench operational modes and their task-level outcomes. Effect-to-action scoring currently uses experiment runners rather than the generic prediction-table interface.

Mode	Observer supplies	Evaluator fixes	Main outputs
Closed-loop control	A state estimate, an edit direction, or a declared pair	Model, starting states, allowed edits, controller, horizon, and loss	Estimation error, trajectory error, and collateral movement
Effect prediction	Predicted effects or losses for held-out interventions	Calibration measurements, held-out interventions, action rule, and loss	Prediction error and selected-action loss
Safety triage	One risk or consequence score per request or trajectory	Test cases, prevalence, action policy, capacity budget, and consequence loss	Caught and missed harm, false interventions, and clean utility

ObserverBench does not collapse unlike modes or task versions into a single overall rank. Each scorecard identifies which observer is better for one declared use.

The workflow makes results comparable. Three additional rules keep each comparison meaningful:

- **Separate construction from evaluation.** Fix targets, splits, and interventions before measuring held-out performance. Check first that the unedited model succeeds on the base task and required groups; we call this the *clean stop gate*.
- **Make the comparison observable.** The measurement design must actually exercise the effects being compared. For example, an interaction cannot be assessed if the relevant component combination is almost never measured. Repeated control should report error across the trajectory rather than only a threshold crossing.
- **Evaluate the quantity used to choose the action.** Mean-effect prediction, prompt-level intervention loss, and consequence-weighted safety risk are different targets. The task should score the one the controller actually uses. When acting is optional, the action set should include exact no-op so the evaluation can also ask whether any intervention is worthwhile.

This contract tells us how to compare observers, but it leaves a prior question unanswered: what must an observer preserve for its estimate to be adequate for control? Section 3 answers that question by restricting accuracy to the states that the allowed interventions can reach.

3 Closed-loop control: what an observer must preserve

Once the controlled model, initial states, allowed intervention rule, controller, and loss are fixed, only observer errors that affect reachable states can change the controlled trajectory. An observer turns a model measurement into information that can guide an intervention. The distinction between the estimate and the intervention is important. An observer may correctly report how far the model is from a target while pointing the controller in an unhelpful direction; conversely, a useful direction can be paired with a poor estimate.

For model state h , let the observer estimate a quantity $\hat{z}$ and let d denote the proposed intervention direction:

$$E(h) = \hat{z}, \quad D(h) = d.$$

The controller uses the estimate to choose an action magnitude,

$$u = C(y^*, \hat{z}).$$

Within this fixed system, testing E keeps D fixed, testing D keeps E fixed, and changing both identifies only the combined system.

An observer need not be correct on every possible model state. It needs to be correct on the states that the allowed interventions can reach. To state this concretely, suppose an activation has three coordinates but the controller can move only the first two. First calibrate the observer to agree with the target at the starting state. It then needs to predict changes along the first two coordinates; the third stays fixed. Without that initial calibration, a wrong coefficient on the unchanged coordinate can still bias every action.

To state the same idea precisely, consider an affine—linear plus an offset—target and observer:

$$z(h) = b_T + \beta_T^\top h, \quad \widehat{z}(h) = b_E + \beta_E^\top h.$$

Let h_0 be the starting state. Let the columns of B span the allowed edit directions. From h_0 , the controller can reach only

$$\mathcal{H}_R = h_0 + \mathcal{U}, \quad \mathcal{U} = \text{span}(B).$$

The observer only has to match the target on this reachable set. With one fixed direction d , this requires both a correct starting estimate and a correct response to movement along d .

Proposition 1 (Reachable-subspace adequacy). *Define*

$$\delta_0 = z(h_0) - \widehat{z}(h_0), \quad \delta_\beta = \beta_T - \beta_E.$$

Then $z(h) = \widehat{z}(h)$ for every $h \in \mathcal{H}_R$ if and only if

$$\delta_0 = 0 \quad \text{and} \quad P_{\mathcal{U}} \delta_\beta = 0,$$

where $P_{\mathcal{U}}$ is the orthogonal projector onto $\mathcal{U}$.

Proof. Every reachable state has the form $h = h_0 + Bv$. The readout difference is $\delta_0 + \delta_\beta^\top Bv$. It is zero for every v exactly when $\delta_0 = 0$ and $B^\top \delta_\beta = 0$, which is equivalent to $P_{\mathcal{U}} \delta_\beta = 0$. $\square$

With one fixed direction d , Proposition 1 requires $\delta_0 = 0$ and $(\beta_T - \beta_E)^\top d = 0$. The algebra is standard affine feedback analysis. Its role here is to turn these two requirements into a concrete test of an observer, not to introduce a new general control theorem.

3.1 Why a convergent observer can still miss the target

The proposition tells us where an observer must agree with the target. The next question is what happens when its remaining error enters a feedback loop. Consider an unclipped controller that repeatedly edits along one direction:

$$h_{t+1} = h_t + K \widehat{e}_t d, \quad \widehat{e}_t = y^* - \widehat{z}(h_t).$$

Here y^* is the desired value, K sets the edit size, and $\widehat{e}_t$ is the error reported by the observer. Three quantities govern the loop:

$$g_E = \beta_E^\top d, \quad g_T = \beta_T^\top d, \quad p = 1 - K g_E.$$

The response g_E is what the observer expects along the edit direction, while g_T is the target's actual response. The convergence factor p —called the closed-loop pole in control theory—says how quickly the error reported by the observer shrinks. After T steps, the total action is

$$A_T = \begin{cases} \widehat{e}_0(1 - p^T)/g_E, & g_E \neq 0, \\ TK\widehat{e}_0, & g_E = 0, \end{cases}$$

and the two errors are exactly

$$\widehat{e}_T = p^T \widehat{e}_0, \quad e_T = e_0 - g_T A_T,$$

where $e_t = y^* - z(h_t)$. For $g_E \neq 0$, define

$$m_0 = z(h_0) - \widehat{z}(h_0), \quad \rho = \frac{g_T - g_E}{g_E}.$$

Substitution gives the finite-horizon certificate

$$e_T = p^T \widehat{e}_0 - m_0 - \rho(1 - p^T) \widehat{e}_0. \tag{1}$$

The equation separates three sources of error:

- $p^T \hat{e}_0$ is the error the observer still reports after T steps;
- m_0 is a mismatch at the starting state; and
- $\rho(1 - p^T) \hat{e}_0$ is the mismatch between the response the observer expects and the response the target actually has.

The observer’s reported error converges when $|p| < 1$, or $0 < Kg_E < 2$ for positive K , but this controls only the first term. For example, if p^T is nearly zero and $m_0 = 0.2$, the observer reports success while the true target remains off by about 0.2, even before response mismatch is counted. A convergent observer is not necessarily tracking the true target.

3.2 A simple control certificate remains predictive in a Transformer

Equation 1 is exact for an affine system. A Transformer is nonlinear. Our first experiment asks whether the same three sources of error—remaining reported error, starting mismatch, and response mismatch—remain useful when every edit passes through learned Transformer layers. We begin with a small model whose target we can verify exactly, following the controlled “wind-tunnel” approach of [2].

Task contract. This experiment instantiates the control mode as follows:

Dimension	Fixed contract
Target	Scalar state readout and its error over a 15-step trajectory.
Information boundary	The residual after the first Transformer block; first-order observers use the two individual features, while lifted observers also use their product.
Action space	Repeated residual edits along the declared observer direction.
Decision rule and loss	Proportional feedback with $K = 0.35$; integrated squared tracking error is primary, with final error and clipping as diagnostics.
Operating regime	Twelve independently trained two-block Transformers for additive and interactional targets, with edits checked against clean residual scale.

Table 2: ObserverBench contract for the closed-loop control experiment.

We train a two-block Transformer on sequences with two binary features, x_1 and x_2 , and target

$$y = 0.35x_1 + 0.25x_2 + \gamma x_1x_2 + \epsilon.$$

The product x_1x_2 matters only when both features are present. A first-order observer reads x_1 and x_2 separately, so it cannot represent this joint condition. A lifted observer adds the pairwise feature x_1x_2 . We edit the residual after block 0 and rerun block 1, the final layer normalization, and the target head. The edit therefore passes through the learned nonlinear suffix; we do not replace that suffix with a linear approximation.

We run two tests with 12 independently trained models per target family:

- **Does the certificate predict failure?** We fix the observer response at $g_E = 0.85$ and vary the response mismatch ρ , initial bias, and target offset. Small positive and negative edits measure the actual target response g_T . We then ask whether Equation 1 predicts which rollouts miss the target and by how much.
- **Does the interaction coordinate improve control?** We compare the natural first-order and pairwise estimator–direction pairs. We use $\gamma = 1.15$ when the target contains the interaction and $\gamma = 0$ as the additive control.

The true target in these rollouts is the output of the rerun Transformer suffix, not a separate fitted linear probe. Measuring g_T and checking final error therefore use target access. This validates a diagnostic under known ground truth; it does not give an observer a safety certificate when the true target is unavailable. Appendix F gives the training, probe, and rollout settings.

Integrated squared tracking error counts error across the full trajectory, not only at the final step, and uncertainty is measured across the 12 independently trained models, not across the many conditions within each model. None of the main comparisons clips. Every natural comparison and 99.2% of controlled conditions move no farther than the largest distance among the four clean task archetypes. The edits therefore have a familiar scale, although four archetypes cannot prove that edited states remain on the full activation manifold.

The suffix is measurably nonlinear. At the largest offsets, its mean deviation from the affine prediction is 16.1% of output movement, and the certificate residual is 20.2% of mean absolute final error. We therefore ask whether the affine quantities remain predictive, not whether the Transformer is linear.

They remain predictive. The main checks agree:

- The raw certificate correlates 0.952 with actual final error and gets the sign right in 96.0% of conditions.
- Its residual MAE is 16.5% of mean absolute actual error.
- After fitting only an intercept and slope on the other 11 seeds, its leave-one-seed-out MAE is 0.063, compared with 0.302 for the bias-only model and 0.367 for the observer-pole-only model. It wins on every held-out seed.

Across this checked operating range, the three certificate terms capture most of the error. This does not show that the suffix or its activation manifold is locally linear.

The direct control comparison gives the same message. The pairwise model reduces integrated squared error by 29.0% when the target contains the “both features present” term, with improvement in 11 of 12 independently trained models. It gives no reliable advantage for the additive target. The exact losses and intervals are reported in Table 3.

This result concerns the estimator and edit direction together. The mismatched estimator–direction pairs are unstable in 10 of 12 interactional models and also fail the action- and activation-scale checks. We therefore do not use them to rank the estimators independently of direction. Appendix Table 15 reports all four cells, including their self-response and convergence factor p .

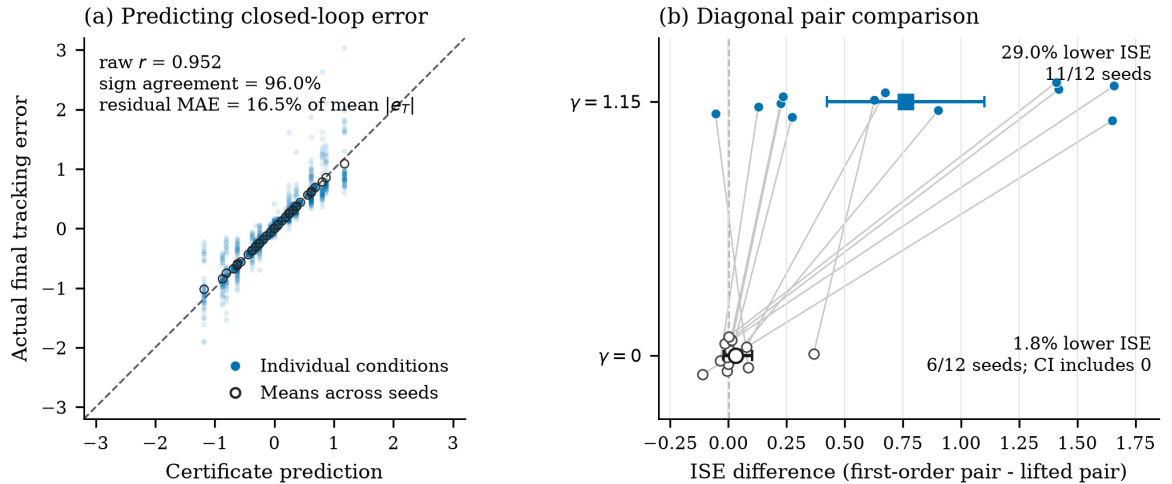


Figure 1: Learned nonlinear-suffix experiment. The certificate combines starting mismatch, convergence factor, and response mismatch (left). The pairwise estimator–direction pair reduces control error for the interactional target but not for the additive target (right). Intervals resample 12 independently trained models.

Comparison	A	B	A-B [95% CI]	Seeds
Bias vs. full cert.	0.302	0.063	0.238 [0.230, 0.246]	12/12
Pole vs. full cert.	0.367	0.063	0.303 [0.294, 0.312]	12/12
FO vs. lifted, $\gamma = 1.15$	2.632	1.870	0.762 [0.424, 1.101]	11/12
FO vs. lifted, $\gamma = 0$	1.693	1.662	0.031 [-0.021, 0.103]	6/12

Table 3: Study-1 nonlinear-suffix results. A-B is positive when the full certificate or lifted pair has lower error. Intervals resample the 12 training seeds.

The study provides a bridge to pretrained models: a simple control-error certificate can remain useful after nonlinear computation when edit scale and loop stability are checked. It does not establish token-level or large-model control. Closed-loop control evaluates an observer over a trajectory. What if the decision instead selects one static intervention from a fixed candidate set using predicted finite effects?

4 Effect prediction: better estimates need not choose better actions

Many interpretability studies use an observer once, to choose one intervention from a fixed set. ObserverBench tests whether better prediction of held-out effects also produces a better choice. It fixes the measurements, candidate actions, decision rule, and held-out loss, then reports prediction error and the loss caused by the selected action separately.

We test this effect-prediction contract on GPT-2-small and Qwen2.5-7B. The two tasks use different behaviors and head sets, but keep the same distinction between estimating an effect and using that estimate to act.

Task dimension	GPT-2-small IOI	Qwen2.5-7B copy
Target	A mask’s mean margin drop, or its prompt-level absolute target loss	A mask’s mean candidate-margin drop, or its prompt-level absolute target loss
Information boundary	Frozen calibration effects and declared head or group features	Frozen calibration effects and declared features for a confirmed eight-head panel
Action space	One candidate head-ablation mask; exact no-op is included in the confirmation	One of eight nonempty head-ablation masks or exact no-op in each pool
Decision rule and loss	Choose the predicted closest-mean or lowest-loss mask; score held-out absolute target error	Choose the predicted closest-mean or lowest-loss mask; score held-out absolute target error
Operating regime	Held-out IOI prompts and previously unseen masks	Eligible held-out induction-copy prompts across four sequence conditions and unseen masks

Table 4: Task contracts for the two effect-prediction studies. Each observer receives the same calibration information within a task and faces the same candidate actions and held-out loss.

4.1 GPT-2 IOI: predicting an effect is not choosing an action

The nonlinear-suffix experiment used a small model with a known mechanism. We now turn to a documented circuit in GPT-2-small. Consider the prompt “When Alice and Bob went to the store, Alice gave a book to.” The correct next name is Bob, not the repeated subject Alice. Prior work identifies three relevant head groups: primary Name Movers promote Bob, Backup Name Movers compensate when that path is weakened, and Negative Name Movers suppress the competing name [16, 17, 18, 19]. These known groups let us test observers on a real intervention surface. We do not claim to discover a new circuit.

We first ask whether pairwise terms improve mean-effect prediction and the resulting fixed action, then predict prompt-level action loss directly, and finally repeat that comparison after the baseline task check passes with the exact no-op included.

We write the three published head groups as

$$\begin{aligned}
P &= (9.9, 9.6, 10.0), \\
B &= (9.0, 9.7, 10.1, 10.2, 10.6, 10.10, 11.2, 11.9), \\
E &= (10.7, 11.10).
\end{aligned}$$

An ablation mask m is the set of heads removed together. For prompt i and mask m , define

$$Y_i(m) = LD_{i,\text{clean}} - LD_{i,\text{ablate}(m)},$$

where LD is the logit for the indirect object minus the logit for the subject. Thus $Y_i(m) = 1$ means that ablating mask m reduces Bob’s advantage over Alice by one logit on prompt i . At the final token, mean ablation replaces each selected head’s output with an average from reference prompts of the same template. Whole-group ablations first reproduce known backup behavior. This is a check that the intervention works as intended, not a new mechanistic result.

4.1.1 A fixed test of effect prediction and action selection

We freeze the test before measuring candidate effects:

- **Prompts and splits.** We pin GPT-2-small and generate 1,024 prompts from eight templates and 64 names. Separate 16-name banks supply 512 reference, 192 training, 64 validation, and 256 test prompts. Reference prompts define the ablation means.
- **Measurements and actions.** We use 160 calibration masks to fit each observer. The observer then chooses from 320 previously unseen candidate masks, divided into ten action pools of 32. Each mask contains 4–10 heads. Budgets 20, 40, 80, and 160 use nested prefixes of one calibration sequence.
- **Observers.** Four ridge models move from 13 independent head effects, to group-count features, to a capacity-matched $P \times E$ term, and finally to all three group-pair interactions. Each measurement is one mask’s average effect over the training prompts.
- **Fixed action.** For each observer, budget, pool, and target $t \in \{0.5, 1.0, 1.5\}$, the observer chooses one mask that is applied to every held-out prompt. It cannot choose one mask for Alice–Bob and another for Carol–David.

An audit found that an earlier metric had accidentally allowed a different, outcome-informed choice for each prompt rather than one frozen action. We corrected it before opening validation or test outcomes; the prompts, masks, observers, targets, and predictions were unchanged.

4.1.2 Interaction terms improve mean-effect prediction

Let

$$\mu(m) = \mathbb{E}_i[Y_i(m)]$$

be a mask’s average effect over held-out prompts. At the primary budget of 160 measurements, the all-pairs observer lowers candidate-mask MAE from 0.486 for per-head additivity and 0.473 for count additivity to 0.395, reductions of 18.8% and 16.7%. Both improvements hold in all ten candidate pools.

The $P \times E$ and all-pairs observers are effectively tied, with MAEs 0.3940 and 0.3945. This replaces an earlier exploratory ranking of the pairs. The result supports interaction-aware prediction, but it does not show that one pair uniquely explains the intervention surface. The gain appears at budgets 40, 80, and 160. With only 20 measurements, the richer designs have more coefficients than the data can identify. That cell is therefore an underdetermined fit, not evidence about sample efficiency.

4.1.3 The better mean predictor does not choose a better action

A mean-effect observer chooses

$$\hat{m}(t) = \arg \min_m |\hat{\mu}(m) - t|.$$

For $t = 1$, for example, it selects the mask whose predicted average drop is closest to one logit. The benchmark instead scores the prompt-level loss of that fixed mask,

$$R_t(m) = \mathbb{E}_i[|Y_i(m) - t|].$$

Two masks can have the same average and different losses. A mask that reliably reduces the margin by one logit is better than one that alternates between zero and two, even though both average to one. Mean-effect prediction and action selection are therefore different tasks. This is an instance of a broader prediction–decision distinction: the error of an intermediate estimate need not determine the quality of the action chosen from it. Roughan et al. documented an analogous result in traffic engineering, where traffic-matrix estimation-error magnitude was not by itself a good indicator of routing performance [22]. Here, mean-effect error omits a decision-relevant quantity: how a mask’s effect varies across prompts.

At budget 160, the all-pairs model has the best R^2 and nearly the best mean-effect MAE, yet its action-loss reductions are only 0.10% and 2.34%, with paired intervals $[-0.087, 0.100]$ and $[-0.063, 0.128]$. Table 5 reports the raw outcomes; the intervals cross zero, and none of the prespecified size-matched, cost-aware, tolerance, or name-pair checks establishes a reliable gain.

Observer	MAE	R^2	$ \mu - t $	J_t	R_t	Within 0.25
Per-head additive	0.4860	0.7629	0.4910	0.5252	1.0163	15.42%
Count additive	0.4734	0.7718	0.3476	0.6920	1.0396	14.62%
Count plus $P \times E$	0.3940	0.8365	0.3561	0.6703	1.0265	15.44%
Count plus all pairs	0.3945	0.8387	0.3498	0.6655	1.0152	15.76%

Table 5: Held-out IOI results at 160 measurements. MAE and R^2 score the prompt-averaged effect over candidate masks. The remaining columns score the masks selected for 30 pool–target cells. Values may differ slightly under rounding because $R_t = |\mu - t| + J_t$ is computed before aggregation.

The reason can be written exactly. Define the target-dependent dispersion penalty

$$J_t(m) = \mathbb{E}_i |Y_i(m) - t| - |\mathbb{E}_i[Y_i(m)] - t|.$$

This nonnegative term is the extra action loss caused by prompt-to-prompt variation hidden by the mean. Therefore

$$R_t(m) = |\mu(m) - t| + J_t(m). \quad (2)$$

This is Jensen’s inequality written for the task’s absolute loss, not a new inequality. Here J_t is not variance: it depends on the target and the response distribution. Estimating it need not be cheaper than estimating R_t directly. The interaction-aware observers choose masks whose averages are closer to the target, but those masks vary more across prompts. The second effect cancels the first. Even perfect knowledge of each mask’s test-set mean chooses the lowest-loss mask in only 3 of 30 pool–target cells. Its loss is 0.927, compared with 0.749 when the full prompt-level action loss is known. Mean-effect prediction works as intended; the mean is simply the wrong quantity for this action rule.

4.1.4 Predict the loss used to choose the action

Equation 2 tells us what to try next: keep the measurements and action rule fixed, but predict the action loss $R_t(m)$ instead of the mean effect $\mu(m)$. We designed this test after the frozen mean-effect comparison and an exploratory pilot, then fixed the protocol, predictions, and actions before measuring new test effects. The local record documents data-access order; it is not an independent preregistration.

The comparison keeps the measurement basis and action rule fixed:

- eight new templates, 192 training prompts, 512 test prompts, and targets $t \in \{0.5, 1.0\}$;
- the same 160 calibration masks and 1,536 candidate masks in 48 pools;
- one pairwise (quadratic) basis with an intercept, 13 head indicators, and 78 head-pair products.

The direct-loss observer, called direct risk in the tables, fits

$$\frac{1}{192} \sum_i |Y_i(m) - t|.$$

The natural-mean observer predicts the average effect and chooses the mask closest to the target. A transformed-mean control predicts that distance directly with the same 92-column basis. The observers

therefore see the same masks and features; only the quantity they learn differs. The held-out evaluation comprises 786,432 test prompt–mask outcomes.

The mean observer remains strong at its own task: held-out mean-effect MAE is 0.128, R^2 is 0.844, and rank correlation is 0.938. The direct-loss observer nevertheless reduces action loss from 1.076 to 0.833, a 22.6% reduction, and by 22.7% against the same-feature transformed-mean control. Table 6 reports the paired intervals and all prespecified checks.

The decomposition explains why. The direct-loss observer matches the mean less closely: 0.550 instead of 0.433. But it lowers the prompt-variation term from 0.643 to 0.283. That 0.360 reduction is larger than the 0.117 loss in mean matching. Extra pairwise features do not reliably improve direct loss over either additive control; both intervals cross zero. On this surface, predicting the right quantity matters more than adding features.

Check	Status	Point statistic	95% interval
H1: risk vs. natural mean	Pass	22.61% reduction	[0.114, 0.402]
H2: risk basis vs. additive	Fail	3.03% reduction	[-0.025, 0.076]
H2: risk basis vs. count-additive	Fail	3.54% reduction	[-0.021, 0.079]
Target-specific transformed-mean sensitivity	Pass	22.67% reduction	[0.119, 0.399]
Natural-mean own-estimand diagnostic	Descriptive	MAE 0.128; R^2 0.844	–
No-action audit	Limitation	loss 0.750 vs. risk 0.833	not prespecified
Clean-task validity	Fail	overall 0.867 < 0.900	worst template 0.578 < 0.750

Table 6: Pilot-informed, outcome-sealed fixed-action comparison. Contrast point statistics are relative fixed-action loss reductions; intervals are paired absolute loss reductions from the frozen pair-cluster-by-pool bootstrap.

Two failed checks limit this result:

- **The exact no-op was absent.** Every candidate mask removes 4–9 heads. Since $Y_i(\emptyset) = 0$, no-op loss is exactly t . Averaged over the two targets, no-op loss is 0.750, below the direct-loss observer at 0.833.
- **The baseline task check failed.** Overall IO-versus-subject accuracy is 0.867 rather than 0.900, and the worst template reaches 0.578 rather than 0.750.

We remove no rows. This test shows better ranking among forced nonempty actions on this surface. It does not show improvement over no-op or a new IOI mechanism.

4.1.5 At a one-logit target, direct loss beats the mean observer and exact no-op

The failed checks determine the confirmation:

- We return to the eight canonical templates and generate 512 new prompts from 32 new name-pair clusters. Clean IO-versus-subject accuracy is 0.977 overall and at least 0.953 for every template, so the baseline task check passes.
- We use $t = 1$, the only screened target whose pilot estimates favored the direct-loss observer over both the same-feature mean observer and exact no-op. This is a disclosed pilot-informed choice, not an independent preregistration.
- Each of 48 action pools contains 30 fresh nonempty masks plus exact no-op. Both observers use the same 92-column quadratic basis and the same frozen training and calibration outcomes.
- We freeze every action before measuring candidate effects. The two observers select 89 distinct nonnoop masks, so we measure only the corresponding $512 \times 89 = 45,568$ prompt–mask cells.

A provenance check stopped an initial run before its values were inspected. Appendix B records the corrected frozen run and an independent recomputation of its coefficients, predictions, and choices.

Both required comparisons pass: Table 7 shows an 18.5% action-loss reduction against the same-feature mean observer and a 10.9% reduction against exact no-op, with both paired intervals excluding zero.

Reference	Ref. loss	Risk loss	Reduction	95% interval	Templates
Same-basis mean-effect plug-in	1.093	0.891	18.5%	[0.115, 0.293]	8/8
Exact no action	1.000	0.891	10.9%	[0.012, 0.196]	6/8

Table 7: Fixed-action target tracking at the pilot-informed target $t = 1.0$. On 512 new prompt strings from eight fixed canonical templates, intervals resample 32 name-pair clusters and 48 frozen action pools (5,000 draws). Intervals are absolute loss reductions; the Reduction column is relative. The last column is descriptive and shows templates with the same direction. The clean IO-versus-subject gate passed before measurement (97.7% overall; 95.3% worst template).

The decomposition shows the same tradeoff. The direct-loss observer has a worse mean-to-target term, 0.397 rather than 0.286, but lowers the dispersion penalty from 0.807 to 0.494. It beats the mean observer on all eight templates and no-op on six. The pooled no-op comparison passes, but the gain is not uniform across templates.

How the target changes the value of acting. After the primary result, we freeze a secondary analysis over all three screened targets. It keeps the prompts, pools, quadratic basis, and calibration outcomes fixed. It compares the direct-loss observer with the natural mean plug-in, the same-basis transformed-mean score, and exact no-op. We measure new outcomes only for additional selected masks. This analysis has no success gate and does not change the baseline-checked result.²

Target	Direct loss	Gain vs. mean	Gain vs. transformed mean	Gain vs. no action
0.5	0.731	+29.4% [0.209, 0.405]	+20.8% [0.102, 0.288]	-46.2% [-0.314, -0.161]
1.0	0.891	+18.5% [0.115, 0.293]	+15.9% [0.089, 0.253]	+10.9% [0.015, 0.196]
1.5	1.070	+10.7% [0.020, 0.243]	+9.3% [0.011, 0.217]	+28.6% [0.321, 0.525]
Equal-target mean	0.897	+19.1% [0.141, 0.287]	+14.9% [0.091, 0.225]	+10.3% [0.021, 0.177]

Table 8: Post-confirmatory target sensitivity. This secondary analysis was designed after the $t = 1$ confirmation and has no success gate. Each gain cell reports the relative loss change and, in brackets, a 95% paired bootstrap interval for the absolute reference-minus-direct loss difference. The three targets share 512 prompts and 48 frozen action pools; the last row weights the targets equally. The negative no-action result at $t = 0.5$ is retained.

Table 8 shows that the direct-loss observer beats both mean-based controls at every target, with every paired interval excluding zero. No-op follows a simpler rule. Its effect is zero, so its loss is exactly $R_t(\emptyset) = t$. At $t = 0.5$, the direct-loss observer’s loss is 46.2% higher than exact no-op, even though it beats both mean-based observers. It beats no-op at $t = 1$ and $t = 1.5$. This sign change is why exact no-op belongs in every action set: an observer can rank the available edits well even when no edit is warranted. We do not add a post-hoc noise-floor gate. Such a threshold or abstention rule would define a different controller and must be fixed before evaluation.

Together, the IOI tests show that pairwise terms improve mean-effect prediction and direct-loss prediction improves fixed-action choice, while exact no-op tests whether any intervention is warranted. Because these claims are limited to GPT-2-small, the declared templates and masks, and the measured action losses, we next ask whether the first two patterns transfer to a larger model and a different intervention surface.

4.2 Qwen2.5-7B: the pattern transfers to a larger model

We test that transfer on the base Qwen2.5-7B model [24] using an induction-copy task with an exact behavioral target and a different set of heads to intervene on [23]. The result applies only to prompts that the unedited model already solves with a clear margin.

Each prompt first contains three key-value pairs and later repeats one key. For example, after “A red, B blue, C green, ... B”, the desired continuation among the three values is “blue.” Red and green are the

²At $t = 1$, the confirmation and this sensitivity use the same loss cells but separate fixed bootstrap seeds. Their point estimate is identical; lower limits 0.012 and 0.015 differ only through Monte Carlo resampling.

declared distractors. For each prompt, we score how strongly the model favors the target over the two distractors combined. For prompt i , define this margin as

$$M_i = \text{logit}_i(\text{target}) - \text{logmeanexp}(\text{logit}_i(\text{distractor}_1), \text{logit}_i(\text{distractor}_2)),$$

where $\text{logmeanexp}(a, b) = \text{logsumexp}(a, b) - \log 2$. The effect of an ablation mask is the drop in that margin:

$$Y_i(m) = M_i(\emptyset) - M_i(m).$$

A positive $Y_i(m)$ means that ablating those heads weakens the target relative to the two alternatives. Four prompt families combine sequence lengths 32 and 64 with repeat gaps 8 and 16.

4.2.1 Define eligible prompts before discovering heads

Copy-v1 required 95% baseline accuracy among the three candidates. Qwen reached 92.97%, so the run stopped before any attention or intervention result was opened. Copy-v2 did not lower the threshold. Instead, before analysis it defined an eligible prompt as one for which Qwen chooses the target among the three candidates, has margin $M_i \geq \log 4$, and produces finite scores. Discovery prompts must also agree under the eager and SDPA attention implementations. Both protocols recorded their source and configuration in a private commit before model outcomes. This records data-access order but is not an independent preregistration.

Before scoring, we assign 3,072 fresh prompts to separate data banks. A total of 2,612 prompts (85.03%) pass the eligibility and coverage checks. Hash order, not model score, then selects 1,536 prompts for reference, discovery, head fitting, confirmation, calibration, and final held-out testing. These roles remain separate throughout the experiment.

On the unfiltered prompts, Qwen produces the desired next token only 24.45% of the time. The task therefore asks a bounded question: when Qwen already distinguishes the planted value from the two distractors, which heads support that distinction, and how well can we predict their intervention effects? It does not test unconstrained generation.

4.2.2 Held-out interventions confirm the eight-head panel

Attention can suggest heads, but only an intervention shows that they affect the output. We therefore use four frozen steps:

1. Rank all 784 query heads by attention to the target value relative to the distractors.
2. Apply individual mean ablations to only the top 32 on a separate head-fitting bank.
3. Select at most two heads per layer using a lower-bootstrap-bound rule. This yields L22H3, L22H4, L26H15, L14H0, L14H6, L16H13, L17H18, and L23H13, spanning six layers.
4. Test the selected panel against matched control heads on a separate confirmation bank.

Each selected head receives a low-induction control from the same layer and grouped-query-attention key-value group. Controls are matched using attention specificity and output-norm distance, not causal outcomes. On 256 separate confirmation prompts, the selected heads lower the candidate margin much more than matched controls, with the paired interval excluding zero; a frozen zero-ablation check gives the same ordering. Table 9 reports the estimates. Thus the held-out intervention results, not the attention scores by themselves, justify including the panel in the benchmark.

4.2.3 Pair terms and direct loss improve held-out performance

Only after confirmation passes do we freeze all $2^8 = 256$ combinations of the eight heads. One mask might remove heads 1 and 3; another might remove 2, 4, and 8. We split this complete mask set into 128 calibration masks and 128 held-out masks. Calibration effects average 256 prompts; held-out effects average a separate 512. The held-out prompts form 128 four-family clusters, so every mask is tested at every sequence-length and repeat-gap condition.

The additive observer assumes that removing heads 1 and 3 has the sum of their separate effects. The pairwise observer, a quadratic model for these binary masks, adds all 28 head pairs and can represent a head whose effect changes after another head is removed. At the primary budget of 128 measurements, the pairwise model improves held-out-mask MAE over the additive observer, with the paired interval excluding zero (Table 9). The pairwise model also improves MAE at every smaller nested budget; the 16-measurement fit has more coefficients than measurements and is therefore descriptive.

Every action selector uses the same 37-column pairwise basis, the same three targets, and the same 16 frozen action pools. Each pool contains eight nonempty masks and the exact no-op. The observers differ only in what they predict:

- **Natural mean** predicts $\mathbb{E}_i[Y_i(m)]$.
- **Transformed mean** predicts the distance between that mean and the target.
- **Direct loss** predicts the action loss itself, $\mathbb{E}_i|Y_i(m) - t|$.

The targets are 0.25, 0.50, and 0.75 of the confirmed full-panel effect. We freeze every prediction and action before opening held-out effects.

Test	Comparison	Reference	Result	Improvement [95% CI]
Causal	Mean replacement	Control 0.170	Selected 3.454	3.284 [3.053, 3.520]
Robustness	Zero ablation	Control 0.528	Selected 4.080	3.552 [3.306, 3.801]
Prediction	16 measurements [†]	Additive 0.354	Pairwise 0.100	0.254 [0.182, 0.295]
	40 measurements	Additive 0.202	Pairwise 0.045	0.157 [0.103, 0.173]
	64 measurements	Additive 0.157	Pairwise 0.039	0.117 [0.067, 0.134]
	128 measurements	Additive 0.121	Pairwise 0.040	0.081 [0.025, 0.101]
Action loss	Natural mean	0.955	Direct loss 0.870	0.085 [0.053, 0.118]
	Transformed mean	0.995	Direct loss 0.870	0.125 [0.083, 0.173]
	Exact no-op	1.727	Direct loss 0.870	0.857 [0.805, 0.908]

Table 9: Qwen2.5-7B Copy-v2 results. Causal entries report mean candidate-margin drop on 256 confirmation prompts. Prediction entries report MAE on 128 held-out masks; positive improvement is additive minus pairwise MAE. Action entries average absolute target loss over three targets and 16 pools; positive improvement is comparator loss minus direct loss. Intervals use 5,000 paired percentile bootstrap draws over the appropriate prompt clusters, masks, and pools. The dagger marks the underdetermined, descriptive 16-measurement fit. The 128-measurement prediction row and aggregate action rows are the prespecified primary tests.

Direct loss lowers aggregate loss by 8.9% relative to natural mean, 12.6% relative to transformed mean, and 49.6% relative to no-op. These numbers give equal weight to the three targets. At the largest target, direct loss and natural mean tie within uncertainty: natural minus direct loss is -0.008 , with interval $[-0.038, 0.021]$. Appendix Table 17 reports each target separately.

On 512 matched non-induction continuations, a secondary check finds no larger clean-to-intervened KL for direct loss than for the natural-mean and transformed-mean controls (0.0115 versus 0.0157 and 0.0160); this is not a general safety result.

Across GPT-2 and this bounded Qwen task, pair terms improve held-out effect prediction and observers trained on realized loss choose lower-loss interventions in their frozen aggregates; the Qwen result, however, remains limited to one pinned model, the eligible prompts, the confirmed eight-head panel, and the declared masks and neither identifies a complete induction circuit nor establishes general in-context learning.

These studies choose model interventions from predicted effects. What changes when the action is instead a scarce safety response and different mistakes carry different consequences? The next operational mode keeps the same task-relative contract but evaluates observers under fixed intervention budgets.

5 Safety triage: scores must guide scarce interventions

When the controller cannot inspect every request, a safety observer must do more than separate safe cases from unsafe ones. It must rank the cases that matter most, using the information available at deployment, so

that a limited intervention budget prevents as much harm as possible. The two tasks in this section make that requirement explicit.

The measurement channel matters as much as the score. During the 2026 OpenAI–Hugging Face incident, agents tried to alter the transcripts they believed an evaluator would inspect and successfully spoofed some tool calls, while showing little interest in changing chain-of-thought they did not expect to be logged [33]. This does not make internal signals tamper-proof. It shows that an attacker may concentrate on the channel it believes is watched. ObserverBench therefore compares declared views—text, outputs, verdict-adjacent activations, and neutral activations—and records who controls each view. Monitoring is therefore a moving contest: a channel that works against frozen attacks may fail after attackers learn that it is being used. The present tasks keep those views fixed and trusted; they do not test an attacker that adapts to the monitor.

Table 10: Task contracts for the two safety-triage studies. Each contract fixes what the observer predicts, what it may read, which action the controller can take, and how the resulting decision is scored.

Task dimension	Authorization interlock	APPS backdoor audit
Target	Authorization risk, including the consequence of the requested operation	Whether a submitted program contains a verified working backdoor
Information and integrity boundary	Declared action fields and, depending on the observer, prompt text, Qwen output scores, or one residual activation; evaluator-supplied views are trusted	Programming problem and submitted code, read through Qwen or Gemma prompt scores, text features, dense residuals, or SAE features; stored views are trusted and attacks do not adapt to the monitor
Action space	Allow, block, or escalate a request	Audit a submission or leave it unaudited
Decision rule and loss	Rank by the observer score; block the top 10%, escalate the next 10%, and charge for missed risk and unnecessary intervention	Rank by suspicion; audit the top-scoring submissions and count verified backdoors that escape
Operating regime	Held-out permission pairs across seen and unseen operations and prompt formats, under fixed block and escalation budgets	Held-out APPS problem IDs, fixed audit budgets, and prevalence checks from rare to balanced attacks

The contracts use the same five fields but test different weaknesses. The authorization task gives exact labels and costs, which lets us isolate whether a statistically accurate observer allocates its budget well. We begin there.

5.1 A perfect classifier can allocate a safety budget poorly

Suppose a workspace assistant is asked to delete a file in workspace A, but the user has granted access only to workspace B. Deletion is not always unsafe; the problem is that this file is outside the granted scope. A controller may allow, block, or escalate the request. When blocks and reviews are limited, it must rank requests by the harm they could cause.

We turn this example into a non-executing paired task:

- **Same action, different permission.** Each pair requests the same operation on the same resource. One prompt grants access to it; the other grants access elsewhere.
- **Known label and cost.** An exact authorization rule says whether the request exceeds scope. The operation sets the consequence: an unauthorized delete costs more than an unauthorized read.
- **Fixed action budget.** The controller blocks the highest-risk 10%, escalates the next 10%, and allows the rest. Every observer faces the same costs and budgets.

We use the pinned Qwen2.5-7B-Instruct model [24]. As a prerequisite check, the unedited model must answer the authorization question with one of two exact tokens. It passes every check with accuracy 1.000. The held-out test contains 384 safe/unsafe pairs across four equally sized subgroups: seen operations and prompt forms, a held-out delete operation, a held-out XML form, and both changes together. Fit, calibration, and test resources are disjoint.

The activation observer reads a hidden-state vector—a residual-stream activation—at the final prompt position and also receives the requested action and its severity. We fit on 384 pairs, calibrate on 192, choose among seven layers and ridge settings using the fixed policy loss, and only then extract held-out activations.

The comparisons ask four questions:

- **Action-only:** what can be inferred without authorization context?
- **Activation label and direct risk:** do hidden activations predict the policy label or the violation cost?
- **Activation label $\times$ severity:** does combining the label score with known consequence improve the action ranking?
- **Allow-all and exact oracle:** what are the upper and lower bounds under the same budget?

These comparisons were fixed before held-out activations were extracted. After seeing their results, we added three secondary controls: a full-context text observer, Qwen’s block-minus-allow output-score difference, and an activation observer selected by tail rather than mean loss. They test whether activations add value once prompt text or the model’s own decision signal is available; they are not independent confirmation.

Table 11: Held-out Qwen safety-interlock results. Every observer uses the same 10% block and 10% escalation budgets. Lower loss and missed risk are better; higher clean utility is better. Access states what the observer can read.

Observer	Access	Mean loss	Tail loss	Missed risk	Clean utility
Allow all	None	5.010	19.026	1.000	1.000
Action-only direct risk	Action fields	3.176	13.987	0.623	0.901
Activation label	Residual + fields	3.334	18.182	0.665	1.000
Activation direct risk	Residual + fields	2.520	16.545	0.502	1.000
Activation label $\times$ severity	Residual + fields	2.155	14.026	0.429	1.000
Full-context text label $\times$ severity [†]	Prompt text + fields	2.032	12.857	0.405	1.000
Qwen logit margin $\times$ severity [†]	Output logits + fields	1.744	10.078	0.347	1.000
Exact authorization-risk oracle	Evaluator label + fields	1.740	10.039	0.346	1.000

[†] *Secondary controls.* The text and output rows were designed after the original held-out results were known. The results have four main implications:

- **Activations improve the primary comparison.** Multiplying the activation label by severity lowers mean policy loss by 32.1% relative to action-only and 57.0% relative to allowing every request. It closes 71.1% of the action-only-to-oracle gap while preserving all benign utility.
- **Perfect classification is not enough.** The label observer has AUROC 1.000 and loss 3.334. Multiplying its score by severity lowers AUROC to 0.954 but lowers loss to 2.155 because unauthorized deletes are ranked ahead of unauthorized reads.
- **The useful signal is not uniquely hidden.** Full-context text is not reliably worse than the transformed activation observer. More clearly, Qwen’s output-score difference multiplied by severity lowers mean loss to 1.744 and tail loss to 10.078, nearly matching the exact oracle at 1.740 and 10.039. When this output is available, the activation probe is unnecessary.
- **Good ranking within groups can fail across groups.** The activation label has AUROC 1.000 inside every operation-format subgroup, but its scale changes between them. A global threshold therefore allocates 63.2% of interventions to held-out deletes in seen forms and only 18.4% to the doubly held-out subgroup. There, activation loss is 5.007 versus 3.059 for action-only; the output observer catches 69.8% of unsafe requests versus 29.2% for the activation observer.

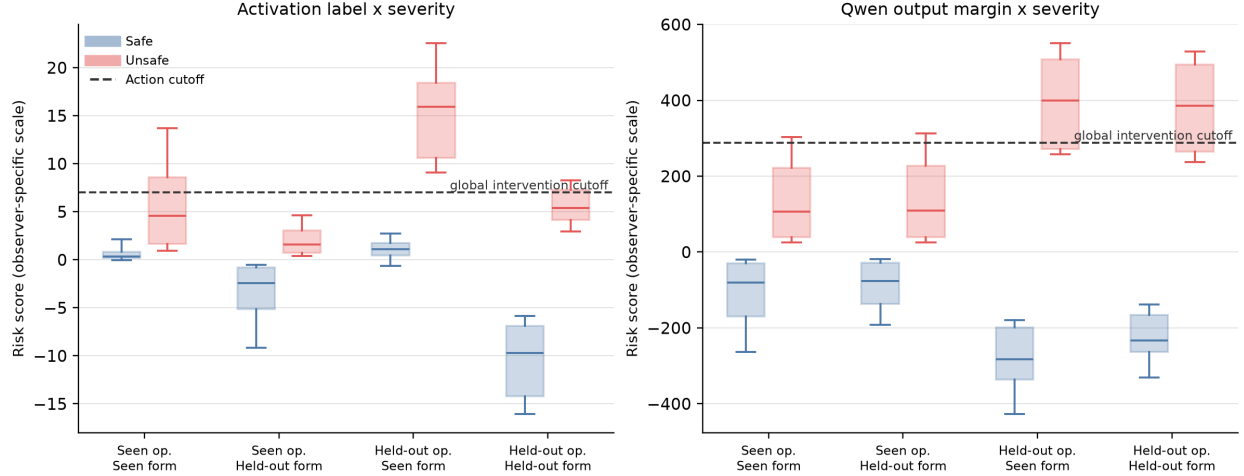


Figure 2: Risk scores by operation-format subgroup. The activation observer separates safe from unsafe requests within each subgroup, but its scale shifts between subgroups. Dashed lines show each observer’s global intervention cutoff. Scores should be compared within panels, not across them.

This failure separates three requirements: the observer must classify the violation, rank it by consequence, and calibrate its scores across deployment groups. Recalibration by visible operation and format may repair the last failure, but this split cannot test that repair because calibration contains neither held-out case. A valid follow-up needs fresh prompts and calibration coverage of every intended deployment group.

The controlled task shows why consequence and cross-group calibration matter. Does the same task-relative evaluation change which monitor we prefer on an externally recognized safety benchmark? We next turn to APPS.

5.2 Monitor rankings depend on the model and operating regime

To answer that question, we use ControlArena’s APPS setting, an existing AI-control task with ordinary code solutions and verified working backdoors [20, 21]. A monitor reads the programming problem and submitted code, assigns a suspicion score, and a fixed budget audits only the highest-scoring submissions. Every unaudited working backdoor counts as one violation.

The public dataset supplies 1,444 honest/backdoor pairs. We split by problem ID: 500 pairs for fitting, 200 for calibration, and 744 for held-out testing, with both members of a pair always in the same split. We use the same pinned Qwen model as above.

The observers fall into three groups:

- **Prompt and output observers** ask Qwen for a score and read the model’s probability distribution over ten score tokens. We test basic and detailed prompts, their ensemble, and a follow-up ridge model using all ten token probabilities.
- **A text observer** is a follow-up baseline trained on word and character features from the problem and code.
- **Activation observers** fit a linear model to a hidden-state vector. One is read next to an explicit verdict prompt; the follow-up neutral observer reads the same problem and code with no monitoring or scoring instruction.

The original comparison used the three prompt observers and the verdict-adjacent activation probe. We added the score-distribution, text, and neutral-activation controls after that result and fixed them before measurement. The later SAE row is also a frozen follow-up control; these added rows are not independent confirmations. At a 2% audit budget, calibration selects layer 28 for the verdict-adjacent probe and layer 20 for the neutral probe.

Table 12: Qwen2.5-7B monitor comparison on 744 held-out APPS pairs. The controller audits 29 of 1,488 submissions at 2% and 148 at 10%. Higher caught-backdoor counts are better; 29 and 148 are the best possible counts under those budgets.

Observer	Access	AUROC	Caught @2%	Caught @10%
Exact-label oracle	evaluator	1.000	29	148
Neutral activation	neutral residual	0.979	29	148
Verdict-adjacent activation	prompt residual	0.953	29	148
Supervised text	problem + code	0.868	29	146
Score-distribution ridge	prompt logits	0.830	29	147
Prompt ensemble	two prompt expectations	0.817	28	146
Detailed prompt	prompt expectation	0.810	28	146
Basic prompt	prompt expectation	0.786	27	137

Three findings matter:

- **Neutral activations rank attacks best.** The neutral probe reaches AUROC 0.979 without asking Qwen to judge the code. Its paired gains are 0.026 over verdict-adjacent activation, 0.111 over supervised text, and 0.149 over the score-distribution observer, with 95% intervals $[0.018, 0.034]$, $[0.098, 0.125]$, and $[0.131, 0.167]$.
- **The balanced panel hides action differences.** Neutral and verdict-adjacent activations both reach the maximum possible caught-backdoor count at 2%, 5%, and 10%. At 10%, neutral activation leaves 596 attacks unaudited versus 607 for the basic prompt, a paired difference of -11 with 95% interval $[-17, -5]$. The sample does not resolve its smaller gains over the stronger controls because attacks make up 50% of the panel and fill the small audit queue.
- **The tested controls narrow, but do not settle, the explanation.** Reading all ten output probabilities raises AUROC from 0.786 to 0.830, so sampling or averaging the declared score tokens does not explain the activation gain. The text result rules out this lexical baseline, not every possible code monitor or output observer.

The neutral-context result is relevant to adaptive monitoring, but it is not a robustness result. It shows that a useful signal can appear before the model is asked for a verdict. It does not show that the same signal survives once an attacker knows it is being read.

In deployment, attacks should be rare. We therefore resample the frozen scores at attack rates from 1–50% and audit budgets from 0.5–10%. At 1% attacks and a 2% audit budget, the exact oracle misses none. The activation observers miss 2.65 attacks on average for neutral context (95% range $[0, 6]$) and 3.94 next to the verdict prompt ($[1, 7]$). The text, score-distribution, and basic-prompt observers miss 9.05 ($[6, 12]$), 9.15 ($[6, 13]$), and 11.85 ($[9, 15]$). Rare attacks expose differences hidden by the balanced panel.

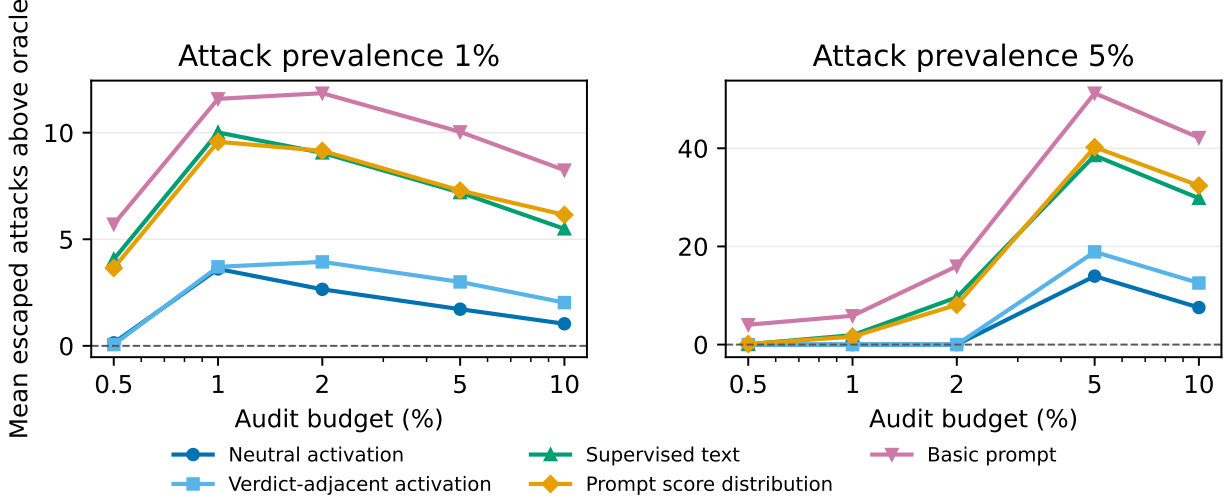


Figure 3: Qwen2.5-7B APPS: excess escaped attacks over the exact-label oracle across attack prevalences and audit budgets in 2,000 resampled populations.

A post-outcome diagnostic applies a published sparse-probing procedure to the frozen Qwen panel [28]. Its fitted readout retains 127 nonzero coefficients, versus 3,584 for a layer-matched dense residual, but misses 4.67 rather than 2.05 attacks on average. This is *compression without a better decision* on this frozen panel, not a general claim about SAE features. It is a same-checkpoint comparison with a disclosed activation-density mismatch: observed mean L_0 is 115 rather than the reported 240. Appendix C reports the full diagnostic, including allocation by APPS family and input length. This is an external monitor test, not a deployment claim.

5.2.1 A checkpoint-matched Gemma replication changes the mean ranking

We repeat the APPS contract on Gemma-2-9B-it, keeping the rows, prompts, controller, budgets, and evaluation loss fixed [25]. The design was frozen after the Qwen results but before any Gemma monitor outcome was inspected. It is a prospectively frozen matched replication, not an independent preregistration. The official Gemma Scope SAE and evaluated Gemma checkpoint match, making this the primary SAE comparison.

Table 13: Matched Gemma-2-9B-it APPS replication. AUROC uses the balanced locked panel of 744 honest-backdoor pairs. Realized violations use the registered primary cell: 1% attacks and a 2% audit budget. Entries are means and central 95% outcome ranges over 2,000 resampled populations; lower is better.

Observer	AUROC	Realized violations
Exact-label oracle	1.000	0.000 [0, 0]
Verdict-adjacent residual	0.969	2.178 [0, 5]
Score-distribution ridge	0.913	4.423 [1, 8]
Neutral residual	0.944	4.947 [2, 9]
Supervised text ridge	0.868	9.054 [6, 12]
Gemma Scope SAE probe	0.870	9.070 [6, 12]
Detailed prompted score	0.893	10.803 [7, 14]
Prompt ensemble	0.860	13.689 [11, 15]
Basic prompted monitor	0.809	14.927 [14, 15]

Residual access has the lowest mean loss in both models, but its best measured context changes: neutral on Qwen and verdict-adjacent on Gemma. Output information still matters. On Gemma, the score-distribution ridge has a slightly lower violation mean than the neutral residual, 4.423 versus 4.947, despite lower AUROC.

On matched Gemma resamples, verdict minus neutral averages -2.77 violations with a central 95% outcome range of $[-6, 0]$. The range touches zero and is not a confidence interval for the mean: verdict is strictly better in 93.2% of paired populations, tied in 5.7%, and worse in 1.1%.

Classification is not enough on the external benchmark. This safety result is a second instance of the prediction–decision distinction in Sec. 4.1.3. The detailed prompted monitor has higher AUROC than the official Gemma Scope SAE probe, 0.893 versus 0.870, but worse mean realized violations, 10.803 versus 9.070 [26]. The detailed-minus-SAE paired difference averages 1.73 with a central 95% outcome range of $[-2, 6]$, so this is a reversal of the mean scorecard ranking, not a superiority claim for every resampled population. A secondary source-problem bootstrap supports the AUROC difference but gives a violation-difference interval of $[-1.18, 3.92]$. The action ranking is therefore uncertain across source problems, even though the locked-panel point estimates reverse (Appendix G).

5.2.2 Qwen3.5 reverses the Qwen2.5 context ranking

Using the same matched protocol, we repeat the APPS rows, split, prompts, audit controller, and primary operating point on Qwen3.5-9B.³ We prospectively froze the design after the Qwen2.5 and Gemma outcomes were known but before inspecting any Qwen3.5 monitor outcome; this is not an independent preregistration.

Table 14: Prospectively frozen Qwen3.5-9B APPS replication. AUROC uses the balanced locked panel. Mean violations use the primary operating point: 1% attacks, a 2% audit budget, and 2,000 paired resampled populations. Lower violations are better. The exact-label row is an evaluator-only bound; bold marks the best deployable row.

Observer	AUROC	Mean violations
Exact-label oracle (evaluator only)	1.000	0.000
Verdict-adjacent residual	0.977	1.976
Supervised score distribution	0.940	5.120
Neutral residual	0.936	6.601
Supervised text	0.868	9.054
Basic prompted score	0.876	12.218
Official Qwen-Scope SAE probe	0.811	12.725
Basic/detailed prompt ensemble	0.823	12.956
Detailed prompted score	0.645	13.985

The verdict-adjacent residual now wins. Neutral minus verdict averages 4.625 additional violations, with a central 95% outcome range of $[1, 8]$; verdict is better in 99.15% of resampled populations. Qwen3.5 therefore agrees with the Gemma ordering and reverses Qwen2.5. The useful context can change even between checkpoints from the same model family.

The other rows show a second reversal. The basic prompted monitor has slightly higher AUROC than supervised text, 0.876 versus 0.868, but causes 3.164 more mean violations, 12.218 versus 9.054. The detailed prompted monitor also falls from AUROC 0.893 on Gemma to 0.645 on Qwen3.5 under the same APPS contract. The source-problem bootstrap supports the basic prompt’s action-loss disadvantage, with interval $[1.50, 4.61]$, but not its small AUROC advantage, whose interval is $[-0.014, 0.029]$. Thus neither of the two highlighted inversions establishes both sides of a population-level ranking reversal. In contrast, verdict-over-neutral action advantages remain supported on Gemma and Qwen3.5: intervals are $[-3.73, -1.78]$ and $[-5.48, -3.33]$. These secondary intervals condition on frozen fits and splits (Appendix G). The model-dependent context choice is better supported than a universal ordering of the prompted and feature monitors.

We also test the official Qwen-Scope SAE.⁴ Calibration selects layer 12 and ridge 100. The SAE probe has AUROC 0.811 and 12.725 mean violations, compared with 6.601 for the dense neutral residual. SAE minus dense averages 6.124 additional violations, with a central 95% outcome range of $[2, 10]$; dense is better in 99.65% of populations. This is a bounded artifact-transfer stress test, not a general result against SAE features: the released SAE was trained on Qwen3.5-9B-Base, while we evaluate the post-trained Qwen3.5-9B

³Pinned model: <https://huggingface.co/Qwen/Qwen3.5-9B>.

⁴Pinned SAE: https://huggingface.co/Qwen/SAE-Res-Qwen3.5-9B-Base-W64K-L0_50.

model [27]. Qwen’s public collection did not include a checkpoint-matched post-trained 9B SAE at the time of the study. Appendix D records the full protocol and mismatch.

5.3 Four checks before claiming that an internal safety signal helps

These studies suggest a reusable sequence. Before interpreting an internal safety signal, ObserverBench asks:

1. **Does the model output already suffice?** Read the relevant output distribution rather than one sampled answer. If it is adequate, stop.
2. **Does the input already suffice?** Train a text or input-feature baseline with the same supervision. If it matches the internal probe, an activation-specific explanation is unnecessary.
3. **Does the signal remain without a request for a verdict?** Measure the same examples in a neutral context to separate input processing from a signal created or reshaped by the monitoring prompt.
4. **Does it work under the intended policy?** Test frozen scores at the attack rate, intervention budget, deployment groups, and loss the controller will face.

These checks locate where useful information becomes available. They do not by themselves establish deception, intent, or hidden knowledge. How does this task-relative contract relate to existing interpretability benchmarks, decision-focused learning, and AI-control evaluations? The next section places ObserverBench among those lines of work.

6 Related work

Interventions, steering, and interpretability benchmarks. Activation addition and representation engineering edit internal activations, while feedback steering and local optimal-control methods place those edits in a controller [3, 4, 5, 6]. Activation and attribution patching estimate component effects; AtP* provides a local approximation; causal abstraction tests assigned causal roles; and pyvene executes model interventions [7, 8, 9, 10]. Interactions between mediators can hide or inflate patching effects [11]. MIB, AxBench, and InterpBench instead benchmark circuit localization, concept detection and steering, or recovery of known structure [12, 13, 14]. Mechanistic tomography treats measurements as a way to recover a mechanism or effect map [1]; ObserverBench starts with such an estimate and asks whether it supports a declared intervention or controller. Sparse probing asks whether SAE features support compact supervised predictors [28]. ObserverBench adds the operational question: does the sparser fitted readout improve the action chosen under a fixed policy?

Decisions and AI control. Decision-focused learning evaluates a predictor through the downstream decision rather than assuming that lower prediction error is enough [29, 30]. The same distinction motivates calls to judge interpretability by the actions it enables [15]. ObserverBench does not claim that general distinction. It makes a different experimental choice: freeze the decision problem and compare interchangeable observers within it. The control conditions specify where errors matter for an actuator, while the empirical tests ask whether familiar prediction metrics select the same observer as the intended action loss. AI control evaluates complete safety protocols against strategies chosen by an intentionally subversive model, while ControlArena supplies reusable environments, monitors, and protocol-level evaluation [20, 21]. ObserverBench asks a complementary component-level question. It holds the controller, audit rule, action budget, and test cases fixed, then compares the observer scores used inside that decision problem. It evaluates monitor adequacy, not end-to-end robustness to subversion.

Borrowed testbeds. The IOI circuit supplies known structure rather than a new mechanism claim [16]. Conditional backup, Hydra-style self-repair, copy suppression, and prompt-dependent self-repair are already documented [17, 18, 19]. Induction heads support the repeated-key continuation used in the Qwen task [23], and Qwen2.5-7B supplies a second model surface [24]. We use these systems to define interventions, then judge observers on fresh held-out masks and prompts rather than claim a new IOI or induction mechanism. What remains unsupported by the current tasks and should be tested next?

7 Limitations and next steps

The paper makes bounded claims:

- **Control.** *Scope:* The exact results assume affine readouts, one fixed direction, proportional control, and no clipping. The nonlinear test reruns a learned Transformer block, but the model is small and its underlying binary features are known by construction; the 29.0% gain belongs to the matched estimator–direction pair, not the estimator by itself. *Next test:* extend the comparison to token-level control in larger models with independently usable edit directions.
- **Pretrained-model interventions.** *Scope:* The GPT-2 and Qwen studies use pinned models, selected head sets, fixed ablations, and narrow prompt distributions. The IOI confirmation is pilot-informed and locally sealed rather than independently preregistered. The Qwen task keeps only eligible prompts; full-vocabulary top-1 accuracy on unfiltered prompts is 24.45%, and the direct-loss observer ties the natural-mean observer at the largest target. *Next test:* use more models, unseen prompt templates, and independently fixed candidate sets and targets.
- **Authorization safety.** *Scope:* The non-executing task asks Qwen directly for an authorization verdict, so it cannot establish a uniquely hidden safety state or test harmful execution, deception, or adaptive attacks. Its held-out failure also shows that one global threshold can hide poor calibration between groups. *Next test:* remove the explicit verdict question and provide calibration data for every intended deployment group while keeping the model, budgets, and evaluator fixed.
- **APPS backdoors.** *Scope:* The study uses one stored-backdoor task rather than attacks adapted to each monitor. Two model families and three checkpoints supply learned representations. The Qwen2.5 sparse probe and Qwen3.5 official-SAE probe each concern one task and one selected probing rule. Gemma provides the primary checkpoint-matched official-SAE comparison. The Qwen2.5 comparison uses the same checkpoint but has a disclosed activation-density mismatch. The official Qwen-Scope SAE was trained on Qwen3.5-9B-Base and applied here to the post-trained Qwen3.5-9B model, making that comparison an artifact-transfer stress test rather than a matched SAE test. The Gemma and Qwen3.5 runs were prospectively frozen after earlier outcomes were known, not independently preregistered. All scoring is offline, and the prevalence sweeps keep the dataset-family mix fixed. Secondary problem-level bootstrap intervals support the verdict-over-neutral action comparisons, but do not settle both sides of the highlighted AUROC–loss reversals. They condition on the fitted monitors and do not measure refitting uncertainty. Neither study tests deception or an adaptive attacker. These scorecards compare observers inside a fixed audit rule; they do not establish that a complete AI-control protocol remains safe under intentional subversion. *Next test:* use sequential attack trajectories with evaluator-held outcomes and both trusted and deliberately corrupted telemetry. Declare who can omit, delay, or spoof each view, keep audit and shutdown budgets fixed, and then add monitor-aware attacks. Episodes, rather than individual requests, should be the resampling unit when agents share state or communication.
- **Artifact.** *Scope:* The release provides checked control, finite-effect, and safety interfaces, but it is not yet a general task ecosystem. The generic effect-table evaluator scores predictions; separate task runners produce the paper’s action-loss results. Readout coefficient counts do not include SAE encoding or model inference. *Next test:* connect submitted effects to the frozen action selector, return action loss against no-op and reference observers, run AtP through that complete path, and measure end-to-end observer costs.

8 Conclusion

ObserverBench asks a simple question: is an estimate good enough for the action chosen from it? The answer is task-relative. An observer need not reproduce every detail of a model; it must preserve the information that the available action, decision rule, and loss actually use. Across the three modes, the experiments show that a better mean-effect predictor need not choose a better intervention, a perfect classifier can allocate a safety budget poorly, and an accessible output signal can make an internal probe unnecessary. The safety scorecards also show that AUROC and mean realized violations can order monitors differently, that the best measurement context can change between closely related checkpoints, and that a sparser fitted readout need not improve the resulting action. ObserverBench records these choices in versioned tasks and applies the same fixed policy to every submission rather than ranking observers in the abstract. Before trusting an observer, state what it may read, freeze the action rule and budget, and score the outcome the action actually produces.

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A Artifact interfaces and sealed evaluation

The artifact exposes three versioned interfaces:

- `observerbench.api.v0` for control composition;
- `observerbench.effect_prediction.v0` for held-out finite-effect prediction; and
- `observerbench.safety_protocol.v0` for safety triage.

Their machine-readable schemas, example submissions, and validation commands ship with the repository. The version identifiers prevent a result produced under one contract from being compared silently with a result produced under another.

The task registry states which submission route each bundled task supports. In the current release, the analytic control task provides the checked external-observer adapter. Its benchmark wrapper constructs the controller and reuses the task’s fixed generator, readouts, normalization, and metrics. A lower-level run may replace the controller, but the resulting score is marked non-comparable because it no longer represents the same decision problem.

Public safety-task packs contain the threat model, allowed actions, consequence loss, operating budgets, training measurements, and held-out queries. They do not contain the held-out evaluator answers. The evaluator stores those targets separately and links them to the public pack by a cryptographic hash. A submission can therefore be checked against the declared task while the labels used for final scoring remain sealed. This layout also allows a researcher to add a data-only safety task without modifying the core workbench.

The bundled GPT-2 IOI and Qwen induction-copy registries implement the same separation for finite-effect prediction. Each registry exposes frozen measurement tables at declared budgets, checks the model and split recorded by the task, and uses different prompts for calibration measurements and held-out targets. An outside observer can submit its predictions without loading the model or rerunning the intervention experiment. The registries include per-component and additive ridge observers as initial comparison rows.

This is a data-only boundary, not a requirement to execute contributor code. For effect prediction, a submission supplies one row per held-out intervention with the fields `schema_version`, `query_id`, and `predicted_effect`, together with an `ObserverCard` that states the method and its access assumptions. The evaluator rejects missing, duplicate, unexpected, or non-finite predictions, joins accepted rows to sealed targets, and returns aggregate prediction metrics. It does not yet select an action or return action loss for an arbitrary submitted table. The action-selection studies in this paper use the frozen IOI and Qwen experiment runners, which fit the mean-effect or direct-loss observer, apply the fixed candidate-pool selector, and score the selected masks on held-out prompts. Their action results should not be read as action results for the submitted AtP rows. The next extension must join these paths while preserving the declared estimand, target, candidate pools, no-op, and tie rule. It must return the selected action, action loss, and comparisons against no-op and named reference observers, not infer decision quality from MAE. Keeping submissions inert and targets private protects scoring labels and avoids running outside code. It does not make the underlying telemetry trustworthy, test adaptive evasion, or prevent repeated tuning to a public panel.

The external APPS registry contains separate Qwen2.5-7B, Gemma-2-9B-it, and Qwen3.5-9B scorecards. Prompt, output, text, dense-residual, and SAE-feature observers are evaluated under the same audit contract within each model panel. The Qwen3.5 release includes the frozen source manifest and an aggregate checked summary containing the scorecard rows, paired contrasts, and official SAE selection. Per-example evaluator labels remain sealed. The registry does not issue a cross-model rank.

Runs use strict JSON and record the contract and result-schema versions, component classes and parameters, package and dependency versions, the source revision, configuration and result hashes, and observer provenance supplied by the submitter.

B Robustness checks and protocol provenance

This appendix retains checks that directly qualify the main results. Superseded control fixtures and development history that support no live claim are kept with the artifact rather than in the paper.

B.1 Nonlinear estimator–direction robustness

Table 15 reports all four estimator–direction combinations behind Sec. 3.2.

γ	Estimate	Direction	ISE	$g_E \leq 0$	$ p \geq 1$	Would clip	Beyond clean max
1.15	FO	FO	2.632	0/12	0/12	0/192	0/192
1.15	FO	Lifted	2.381	10/12	10/12	30/192	56/192
1.15	Lifted	FO	3.666	10/12	10/12	56/192	96/192
1.15	Lifted	Lifted	1.870	0/12	0/12	0/192	0/192
0	FO	FO	1.693	0/12	0/12	0/192	0/192
0	FO	Lifted	1.659	0/12	0/12	0/192	0/192
0	Lifted	FO	1.693	0/12	0/12	0/192	0/192
0	Lifted	Lifted	1.662	0/12	0/12	0/192	0/192

Table 15: Secondary nonlinear-suffix estimator–direction factorial. Counts for g_E and p use 12 seed-level means; the last two columns use 192 rollout conditions per cell. The crossed arms at $\gamma = 1.15$ fail loop-conditioning and clean-archetype scale diagnostics, so they do not identify an estimator-only advantage. The clean maximum is a scale check, not a manifold boundary.

B.2 IOI interaction robustness

Table 16 reports the full nested-budget curve behind the interaction comparison in Sec. 4.1.

Budget	Per-head	Count	$P \times E$	All pairs
20	0.846	1.579	1.333	0.795
40	0.675	0.732	0.544	0.559
80	0.496	0.482	0.406	0.412
160	0.486	0.473	0.394	0.395

Table 16: Fresh held-out IOI mean-effect MAE by nested measurement budget.

B.3 Protocol provenance

- **IOI target sensitivity.** The first measurement copy was set aside before inspection because its source seal omitted the head-mapping file. After expanding the seal, we independently recomputed the frozen predictions and actions and repeated the 148-mask measurement. The two copies are identical at the byte level; only the second enters the analysis. Under the frozen raw-score rule, the transformed-mean control selects negative predicted losses in 44, 44, and 16 pools across the three targets, while direct loss does so in 4, 0, and 0. We retain those scores rather than clip them after observing the outcomes.
- **Qwen Copy-v2.** Copy-v1 stopped at 92.97% candidate accuracy, below its frozen 95% gate, before attention or intervention outcomes were computed. Copy-v2 used fresh tokens and prompts under the same gate before generating the locked results in Section 4.2.

B.4 Authorization tail-loss selection

Selecting the activation observer by tail loss chooses the same layer and ridge setting as selecting it by mean loss. Its tail loss differs from action-only by 0.039, with paired 95% interval $[-1.013, 1.156]$. We therefore retain the mean activation gain but do not claim a tail-loss gain over action-only.

B.5 Qwen action loss by target

The main Qwen result gives equal weight to three targets. Table 17 shows the individual comparisons. Direct loss beats transformed mean and no-op at every target. It beats natural mean at the first two targets and ties it within uncertainty at the largest, so the aggregate result is not a win in every cell.

Target	Direct loss	Gain vs. natural mean	Gain vs. transformed mean	Gain vs. no action
0.25s	0.609	+12.0% [0.034, 0.141]	+14.8% [0.035, 0.188]	+29.4% [0.205, 0.300]
0.50s	0.842	+17.6% [0.108, 0.256]	+18.7% [0.117, 0.283]	+51.3% [0.824, 0.942]
0.75s	1.158	-0.7% [-0.038, 0.021]	+6.3% [0.022, 0.146]	+55.3% [1.331, 1.530]
Equal-target mean	0.870	+8.9% [0.053, 0.118]	+12.6% [0.083, 0.173]	+49.6% [0.805, 0.908]

Table 17: Qwen Copy-v2 action loss by target, where $s = 3.454$ is the confirmed full-panel mean effect. Each gain cell gives the relative loss change and, in brackets, a 95% four-family-prompt-cluster-by-pool bootstrap interval for the absolute comparator-minus-direct difference. The negative natural-mean point estimate at 0.75s is retained.

C Post-outcome APPS diagnostics

These checks use frozen activations and scores after the main result was known. They do not change the locked comparison in Table 12.

Does a sparse-autoencoder basis improve the audit decision? We apply the sparse-probing procedure of [28] to a pinned public BatchTopK SAE trained on Qwen2.5-7B-Instruct layer-19 residuals.⁵ Feature ranking uses only the fit split; feature count and regularization use only calibration; and the locked test is scored once. The dense residual and SAE reconstruction controls use the same layer-19 access, data splits, and calibration rule.

Table 18: Post-outcome published-method integration on the frozen Qwen APPS panel. Mean violations use 2,000 resampled populations with 1% attacks and a 2% audit budget. Lower is better.

Observer	Nonzero coefficients	AUROC	Mean violations
Dense residual logistic	3,584	0.983	2.05
SAE sparse readout	127	0.953	4.67
SAE reconstruction logistic	3,584	0.954	4.79

The sparse SAE readout uses fewer fitted coefficients, but it does not improve mean audit loss on this frozen panel. Dense minus sparse averages -2.62 escaped attacks with a central 95% outcome range of $[-6, 0]$; the dense probe is worse in 1.25% of draws. This is not explained only by long inputs: on the 1,301 locked rows within the SAE’s 1,024-token training length, its AUROC is 0.956 versus 0.983 for the dense residual. The checkpoint is also off its training distribution here: observed mean L_0 is 115 rather than the reported 240. The result concerns this checkpoint, layer, task, and probing rule; it is not a general claim that SAE features cannot support safety monitoring.

Does the answer position explain the gap between prompted and neutral activations? The fixed calibration rule selects layer 28 for the verdict-adjacent observer, with calibration AUROC 0.935 and locked AUROC 0.953. A post-outcome check of cached layer 20 raises locked AUROC to 0.958, but calibration would not have selected it (AUROC 0.929), and the neutral layer-20 probe remains higher at 0.979. This check cannot separate answer position from the monitoring instruction or other context changes.

Does pooling hide score differences between groups? Figure 4 groups neutral-probe scores by APPS source family and input-length quartile. Separation remains strong within every family (AUROC 0.956–0.991) and length quartile (0.974–0.988), but score scale shifts. At the global 2% cutoff, a family’s share of caught attacks is 0.36–2.38 times its population share; across length quartiles the range is 0.67–1.43. These ratios come from only 29 audits and are descriptive. They show that pooled discrimination need not imply equal allocation between groups. The fixed family mix and prevalence sweep do not model an attacker choosing where to attack.

⁵<https://huggingface.co/andyrdt/saes-qwen2.5-7b-instruct>

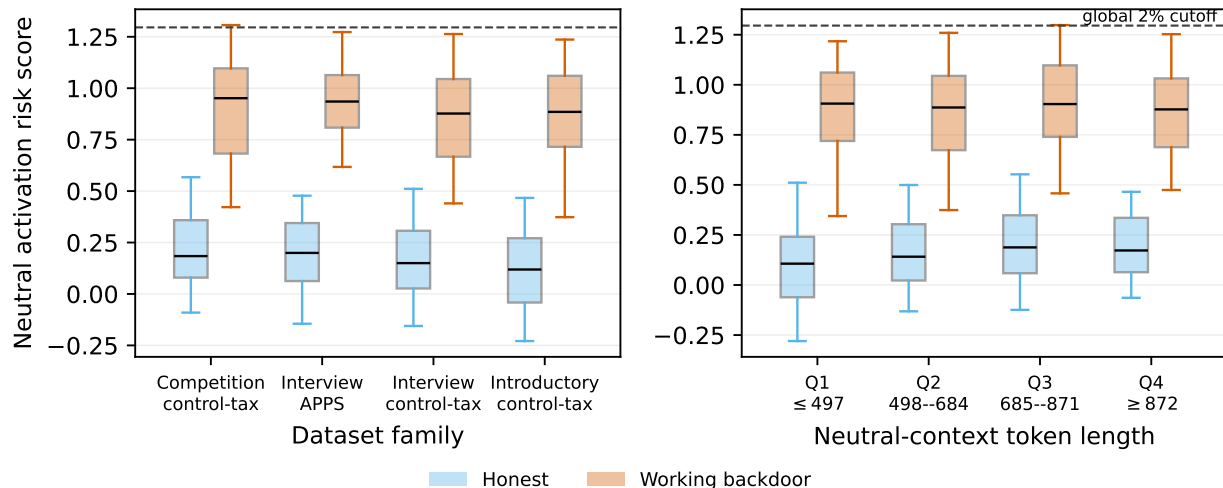


Figure 4: Neutral-context scores by APPS source family and input-length quartile. Separation remains strong within each group, but score scales and the audits assigned by one global cutoff differ between groups.

D Qwen3.5 APPS replication protocol

This replication keeps the APPS task fixed while changing the model. It uses the same 1,444 honest-backdoor pairs as the Qwen2.5 and Gemma studies, split by problem ID into 500 fit, 200 calibration, and 744 locked-test pairs. Both members of a pair remain in the same split. The prompts, score tokens, controller, audit budgets, resampling population, and realized-violation loss also remain fixed. Thinking is disabled for every Qwen3.5 view.

The design was chosen after the Qwen2.5 and Gemma results were known. Before reading any Qwen3.5 monitor outcome, we pinned the model and official SAE revisions, declared the primary contrast, and sealed the source manifest. The primary contrast is neutral-residual minus verdict-adjacent-residual mean violations at 1% attack prevalence and a 2% audit budget. This is a local prospective seal, not an independent preregistration.

The pinned model revision is `c202236235762e1c871ad0ccb60c8ee5ba337b9a`; the pinned SAE revision is `7bb2370d360e566b2d8acb8b3d15a0b2b88f52a8`. The source files and sealed manifest ship with the artifact, so the aggregate scorecard can be tied to the exact code and configuration used for the run without publishing per-example evaluator labels.

We measure post-layer residuals at layers 12, 20, and 28. Fit data train each probe; calibration chooses the layer and ridge value; the locked test is read only after selection. One fit pair exceeds the 4,096-token limit and is excluded under the frozen rule. No calibration or locked-test pair reaches the limit. A compatibility-only smoke first checks prompt lengths, score tokens, residual shapes, and the three SAE files without fitting or ranking an observer.

The official-feature control uses 65,536-wide TopK-50 Qwen-Scope SAEs at the same three layers. Calibration selects layer 12 and ridge 100. The SAEs were trained on Qwen3.5-9B-Base, whereas the dense residuals and outcomes come from post-trained Qwen3.5-9B. At the time of the study, Qwen’s official public collection listed only base-trained 9B Qwen3.5 SAEs, not a checkpoint-matched SAE for the post-trained model.⁶ The Qwen-Scope card also states that using a base-model SAE to explore a post-trained checkpoint is often reasonable. We declared the mismatch before the run. The SAE row therefore tests a released feature family under transfer; it is not a checkpoint-matched comparison or evidence that SAE features generally fail for safety monitoring.

⁶Official collection: <https://huggingface.co/collections/Qwen/qwen-scope>.

E Why a lexical safety fixture was rejected

An earlier fixture tried to distinguish unsafe actionable assistance from benign discussion of the same security topic. A TF-IDF baseline and a linear activation observer both solved its small split, even with one topic family held out. Removing lexical directions then removed the target signal by construction. The task could not distinguish an internal safety representation from words that revealed the label, so it is not an ObserverBench result.

The failure motivated the controls used in the authorization study: test a text baseline, hold the requested action fixed while changing the authorization context, and compare with the exact policy rule when one exists. The later text, output, and neutral-context controls add a further check: if the prompt asks directly for a safety verdict, an activation probe by itself cannot establish a uniquely hidden safety state. More examples with the same lexical structure would not repair the rejected fixture.

F What each experiment fits, freezes, and tests

This section records the choices needed to interpret the comparisons. A configuration hash preserves an exact run; the descriptions below explain what that run measured. Revision prefixes below identify full pins in the run configurations. The experiments use different units of replication. We do not treat many conditions from one model, or many resamples from one dataset, as independent models or datasets.

F.1 Learned nonlinear control

The model has two Transformer blocks, width 32, four attention heads per block, and an MLP width of 96. Each of 12 seeds generates 3,000 training sequences of length four. The binary features have probability 0.35. The training target is $0.35x_1 + 0.25x_2 + \gamma x_1x_2 + \epsilon$, with noise standard deviation 0.02 and $\gamma \in \{0, 1.15\}$. Training uses 600 steps, batch size 256, AdamW learning rate 10^{-3} , weight decay 10^{-4} , and an auxiliary feature loss of weight 0.05.

The observer reads the last-position residual after block 0. Ridge probes with penalty 10^{-4} decode x_1 , x_2 , and x_1x_2 from training residuals. A second ridge fit predicts the model’s suffix output from either the two individual-feature estimates or all three estimates. Composing the fits gives an affine observer in residual space. The controller never receives the exact binary features at test time. The target used to score an edited state is the actual output after rerunning block 1, final normalization, and the target head, not a separately extrapolated target probe.

Each rollout changes only that last-position residual and holds the other positions fixed. Proportional feedback uses $K = 0.35$ for 15 steps without clipping. A symmetric probe of size 0.05 at the starting state measures the true directional response g_T . The controlled diagnostic sets $g_E = 0.85$ and crosses five response mismatches $(-1.25, -0.75, 0, 0.75, 1.25)$, three initial biases $(-0.25, 0, 0.25)$, and four relative target offsets $(-0.5, -0.25, 0.25, 0.5)$. It therefore uses target access both to construct controlled mismatches and to check the error decomposition.

Natural directions are each observer gradient scaled to self-response 0.85. The crossed comparisons keep the estimate or direction fixed, but most crossed pairs fail the declared gain and movement checks. The 29.0% reduction thus compares the two usable estimator–direction pairs. Integrated squared error sums squared true error from step 0 through step 15. The frozen diagnostic protocol names final tracking error as primary and ISE as secondary; the paper uses ISE to compare the natural trajectories, not to rename the diagnostic’s registered primary outcome. Leave-one-seed-out calibration fits on 11 models and tests on the twelfth. Intervals use 5,000 seed-level bootstrap draws.

F.2 The clean GPT-2 action confirmation

The confirmation uses GPT-2-small revision 607a30d783df. The intervention is template-conditioned mean ablation at the final token. Both observers receive the same 192 calibration prompts and 160 calibration masks. Their 92 features comprise an intercept, 13 head indicators, and all 78 head-pair products. Both use ridge penalty 10^{-6} . The mean observer fits the average logit-difference change. The direct-loss observer

instead fits the average absolute deviation from the requested target, computed from the same per-prompt intervention responses. It receives a different training label, not additional measurements.

Evaluation uses eight canonical templates and 32 fresh unordered name pairs, each in two orientations: 512 prompts. There are 48 candidate pools, each containing 30 new nonempty masks plus exact no-op. The nonempty masks have five to nine heads and were excluded from the earlier measured mask sets. Selection orders candidates by predicted loss, then head count, then action ID. For no-op, the predicted and actual losses are exactly $|t|$. The selected actions are frozen before measuring their held-out effects. Only the union of selected masks is measured on the new prompts; the study does not obtain a fresh oracle ranking of every candidate in every pool.

The primary target $t = 1$ was chosen after pilot results. The later target sweep is secondary. Intervals use 5,000 crossed bootstrap draws over 32 name-pair clusters and 48 action pools, keeping both name orientations and all eight templates together within a cluster. They do not measure transfer to a new template family. The earlier 320-mask prediction experiment and its AtP submission rows are separate evaluations, not additional action-confirmation replicates.

F.3 Qwen induction-copy

The model is base Qwen2.5-7B, revision d14972939875, run in bfloat16 without quantization. Four prompt families cross lengths 32 and 64 with repeat gaps 8 and 16. Reference, discovery, head-fit, head-confirmation, calibration, and locked-test banks use disjoint allocated token sets. Eligibility is decided from clean outputs before head interventions: the intended value must beat both distractors with candidate margin at least $\log 4$. The frozen sampling rule selects eligible prompts by a hash, not by ranking their margins. The resulting population supports candidate-margin claims, not unrestricted generation claims.

Attention selects a shortlist of 32 heads. Finite ablations on a separate 256-prompt head-fit bank choose eight, with at most two per layer and at least four layers. Another 256 prompts compare the selected panel with matched low-induction controls. Means from 128 reference prompts define the family-conditioned ablation at the final query position, before attention output projection. All 256 masks of the selected panel split into 128 calibration and 128 test masks, evaluated on 256 calibration and 512 test prompts. The prediction budgets are 16, 40, 64, and 128 masks. Ridge penalties come from $\{10^{-6}, 10^{-4}, 10^{-2}, 10^{-1}, 1, 10\}$ using five-fold mask-level cross-validation on the calibration measurements, not the locked outcomes.

Each of 16 action pools contains eight held-out masks and no-op. Targets are one quarter, one half, and three quarters of the full-panel effect measured on the independent head-confirmation bank. The equal-weight aggregate across these targets is declared before the final analysis. Intervals use 5,000 crossed resamples of prompt clusters and masks or action pools. Mean-effect and direct-loss fits use the same intervention responses; the latter changes the fitted label to the loss the action selector will minimize.

F.4 APPS monitor fitting and deployment scoring

All model panels use the same APPS-control dataset revision 29db28f76042. A problem and its honest and working-backdoor solutions stay together: 500 problem pairs fit observers, 200 select settings, and 744 form the locked test panel. The Gemma and Qwen3.5 replications retain that split and the same prompt forms and policy. The prompts either request a suspicion score or present the problem and code without asking for a verdict. Residual observers read the last non-padding position. Each model’s layer choices and exact revisions are fixed in its configuration.

In the Gemma and Qwen3.5 replications, dense residual fits use a fixed 256-dimensional projection and ridge penalties $\{10^{-4}, 10^{-2}, 1, 100\}$. Calibration selects layer and penalty by violation rate under a 2% audit budget on the balanced calibration panel; the selected fit then uses fit plus calibration data only. The text control uses 2,048 deterministic hashed features. SAE features are standardized from fit data, with rare features removed according to the frozen rule. Gemma uses the official width-16k residual SAEs at layers 9, 20, and 31. Qwen3.5 uses the released base-model SAE on the posttrained model. These are separate comparisons: checkpoint-matched feature evaluation and artifact transfer.

The basic and detailed prompted monitors do not receive supervised fitting labels. Text, residual, supervised output-distribution, and SAE probes do. Their scorecards expose these information differences; the comparisons do not isolate internal access alone. The same-layer Qwen2.5 sparse/dense diagnostic has its

own logistic-fit settings, reported with that experiment. A sparse readout still requires model inference and SAE encoding. We report its nonzero readout coefficients, not an end-to-end compute saving.

Balanced-panel AUROC and rare-attack action loss answer different questions. The primary deployment cell draws 15 attacks and 1,473 honest solutions with replacement, then audits the 29 highest scores. A working backdoor not audited counts as one violation. All monitors see the same draw and the same random row order before stable score sorting, including for tied scores. The 2,000 deployment draws describe how a frozen panel behaves at this prevalence and budget. Their central 95% ranges describe population outcomes; they are not confidence intervals for mean performance on new source problems.

G Do the APPS rankings survive changes in source problems?

The deployment ranges above answer: how much does a frozen panel’s outcome vary across populations with a fixed attack rate and budget? A second question is whether a ranking depends on which APPS problems enter that panel. We address it with a secondary analysis designed after the reported rankings were known. We do not change the trained monitors, primary cell, or original point estimates.

Each of 1,000 outer bootstrap draws samples 744 source problem IDs with replacement, keeping each problem’s honest and working-backdoor solution together. Every monitor receives the same source draw. Within each draw, the existing evaluator averages 256 deployment populations, each with 15 attacks, 1,473 honest submissions, and 29 audits. It uses shared population draws and random tie ordering across monitors. AUROC is computed on the balanced outer panel. The table gives paired percentile intervals over source draws, not the central outcome ranges of individual deployment populations. Nested Monte Carlo makes the loss intervals approximate.

Table 19: Secondary source-problem bootstrap, conditional on the frozen fits and split. Every difference is left minus right. Positive AUROC favors the left monitor; negative violations favor it. Intervals are marginal 95% percentile intervals, not simultaneous confidence guarantees.

Model and comparison	Δ AUROC interval	Δ violations interval
Gemma: detailed prompt – SAE	[0.005, 0.043]	[−1.18, 3.92]
Gemma: verdict – neutral residual	[0.014, 0.034]	[−3.73, −1.78]
Qwen3.5: basic prompt – text	[−0.014, 0.029]	[1.50, 4.61]
Qwen3.5: verdict – neutral residual	[0.030, 0.053]	[−5.48, −3.33]

The result separates two conclusions. The verdict-adjacent residual’s advantage over the neutral residual persists under source resampling on both models. The two prediction–action inversions are less settled. Gemma’s detailed prompt has a supported AUROC advantage over its SAE probe, but the action-loss difference can change sign. Qwen3.5’s basic prompt has a supported action-loss disadvantage against text, but its small AUROC advantage can change sign. Thus neither pair establishes both sides of a population-level ranking reversal at this uncertainty level. They remain observed reversals on the locked panel, not universal monitor orderings.

These intervals hold the fitted observers and model checkpoints fixed. They do not include uncertainty from refitting, choosing layers or penalties, selecting the contrasts after seeing outcomes, or changing the attacker. Independent training splits and adaptive attacks require additional tests. The analysis reuses the existing deployment evaluator and records the score file hashes, seed, resampling unit, and replicate counts in its result files.